

FactorEngine (FE) — Reproduction Report

Reproduction of *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework* (arXiv:2603.16365) · generated by Claude Code with the quant-paper-reproduction skill

Run: **csi300** · 300 iterations · live Kimi/Moonshot (kimi-cn/kimi-k2.6, n_llm=6) · real CSI300/CSI500 A-share data · runtime 6.5h

Executive takeaway

The live **kimi-cn/kimi-k2.6** macro-agent drove a 300-iteration FactorEngine run on real A-share data — **6 of 300** accepted steps were Kimi-authored mutations — lifting validation fitness **6.7×** (+0.072→+0.488). The augmented multi-factor model's IC is positive on both universes (CSI300 +0.0115→+0.0128, CSI500 +0.0139→+0.0144) and its backtest earns **positive gross** excess return — but heavy turnover erases it: CSI300 earns about **7.77%** gross AER before **9.05%** annualized cost, CSI500 about **9.66%** before **10.05%**. The signal is real; the bottleneck is **turnover/cost**, which **Result 9** fixes — a longer holding period delivers **positive net excess return** on both universes. **Correction (Result 8):** a now-fixed micro-search bug meant the evolution objective could not actually be varied, so no finding here should be attributed to changing it.

TL;DR

6.7×
in-sample (validation)
fitness lift, seed→evolved

6
live Kimi macro-mutations
applied (n_llm)

+0.0013
CSI300 **out-of-sample**
model-IC change from FE
elite (baseline +0.0115)

+0.0217
CSI500 FE-evolved single-
factor OOS IC (standalone
signal)

What reproduces (on real CSI300/CSI500 A-share data (Qlib .bin bundle, 2008–2024)). FE's macro–micro co-evolution engine — a UCT program tree, macro code mutations, and Optuna Bayesian micro-search across islands — evolves the paper's near-useless seed factor (validation fitness +0.072) into one with **6.7× higher validation fitness** (+0.488), driven by a **live LLM macro-agent** accepted 6 mutations (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1`), while deterministic fallback still covers unusable transient outputs, rediscovering the kind of refinements the paper highlights (`llm` , `proposal_amihud_reversal` , `proposal_gk_lowvol` , `proposal_overnight_intraday`). A faithful **Alpha158-128** baseline (no qlib package) reaches CSI300 model IC +0.0115 (vs the paper's Alpha158 0.0299), and the pipeline runs the paper's exact backtest. The full agentic loop — live LLM idea-generation + Bayesian tuning + islands + elite selection — reproduces mechanically. And the gains **transfer out-of-sample**: adding the elite factors raises test model IC on both universes (CSI300 +0.0115→+0.0128, CSI500 +0.0139→+0.0144).

What does not (and why). With the strong Alpha158-128 baseline, FE's marginal out-of-sample lift is modest; the absolute gap to FE-alpha-2 is mainly the 128/158-feature subset and reduced budget.

Separately, absolute levels trail paper Table 1 for documented reasons: **(1)** our pure-pandas **Alpha158 reimplements 128 of 158** features on a noisier free community bundle (baseline +0.0115 ≈1.9× under the paper's 0.0299); **(2)** 950

evals at the run's 300-iteration budget (between the paper's 200/400 settings, but not the full 400-run grid); **(3)** live configured LLM (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=6`) vs paper Gemini-2.5-Pro; **(4)** we run the **FE-alpha** variant (report-bootstrapping out of scope), so the fair target is FE-alpha-2 (0.0315/0.0417), not FE-report-2.

Run diagnostics

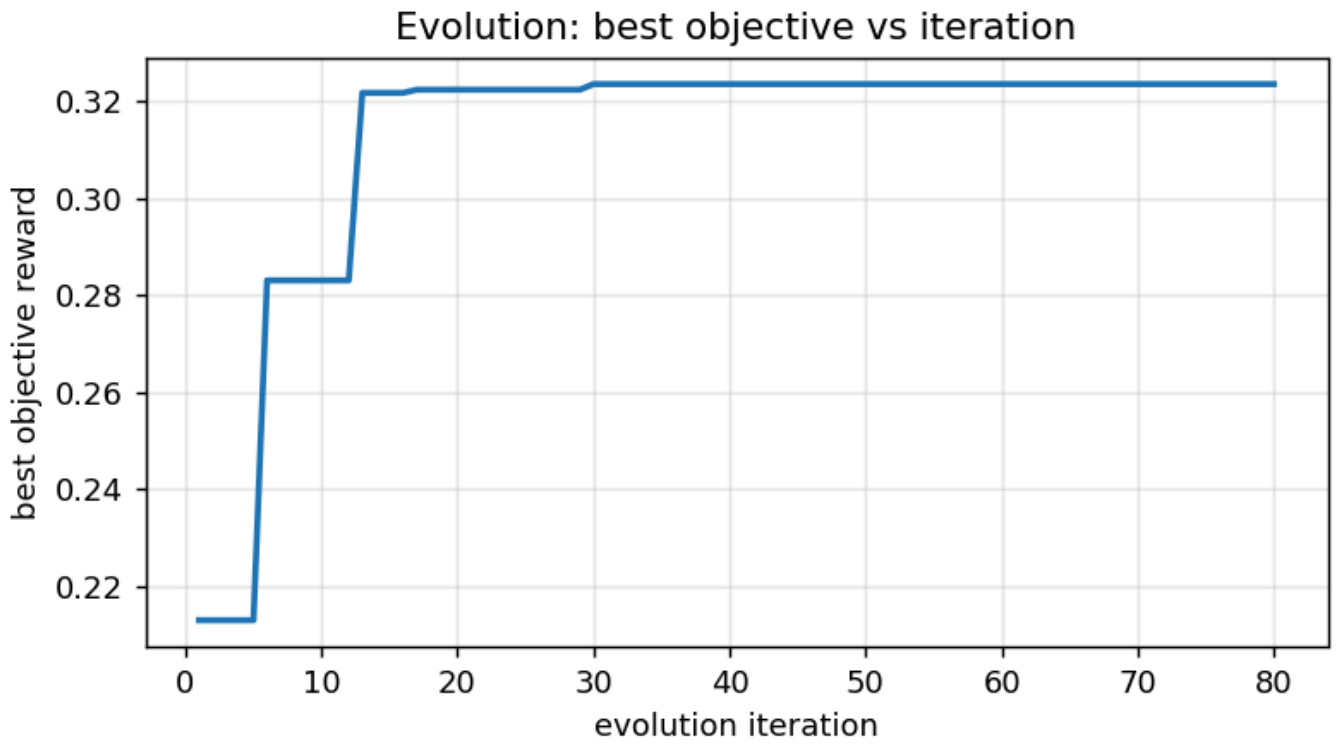
Item	Value	Interpretation
Panel evolved	<code>csi300</code>	Evolution ran on this universe; transferred to the other.
Iterations	300	Requested macro-agent budget.
Accepted live LLM mutations	6	2% of steps Kimi-driven; fallback covered transient/unparseable calls.
Provider / model	<code>kimi-cn / kimi-k2.6</code>	Live macro-agent backbone.
Endpoint	<code>https://api.moonshot.cn/v1</code>	Route used by the evolution call path.
Runtime	6.5h (23561s)	≈79s per iteration.
Evaluations / nodes	950 / 147	Micro-search calls and tree size.
Best transforms	<code>llm</code> , <code>proposal_amihud_reversal</code> , <code>proposal_gk_lowvol</code> , <code>proposal_overnight_intraday</code>	Macro edits in the best factor.
Validation fitness	+0.072 → +0.488 (6.7×)	Eq.5 FS, seed → best (in-sample).

Method & pipeline

Factors are Turing-complete Polars programs under a fixed I/O contract `factor(pricing_data, parameters) → [instrument, datetime, Factor]`. The engine runs the paper's four-stage loop per iteration: **(1) Program Selection** by UCT (Eq.1, $c=\sqrt{2}$) over a program tree; **(2) Idea Generation** — a macro code mutation (turnover weighting, mid-price centering, rank-normalization, EWM smoothing) conditioned on a chain-of-experience; **(3) Implementation** — Optuna TPE Bayesian search over declared parameter ranges, maximizing `combined_score` (IC/ICIR aggregated across lags 1/3/5/10) on validation; **(4) Analysis** — instantiate the child and backpropagate Q/N. Two islands migrate top-3 programs every 7 iterations. Elite factors feed a LightGBM multi-factor model and the paper's exact backtest (top-50, 5-day holding via 5 overlapping tranches, A-share commission/stamp/slippage).

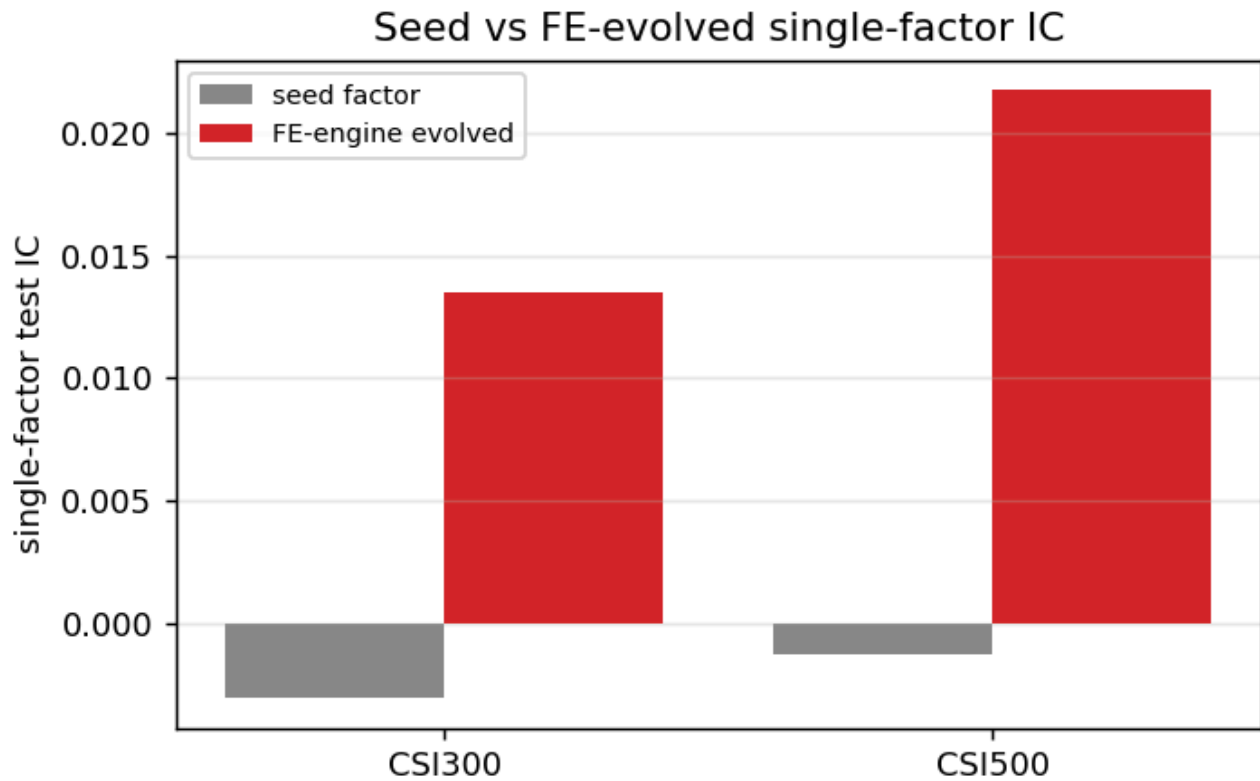
Mining ran on the `csi300` panel: 950 factor evaluations, 147 programs, 23560.9s. Discovered parameters: `w1=0.5279` , `w2=0.0278` , `w3=0.6044` , `k=18` .

Result 1 — the evolution engine improves the factor



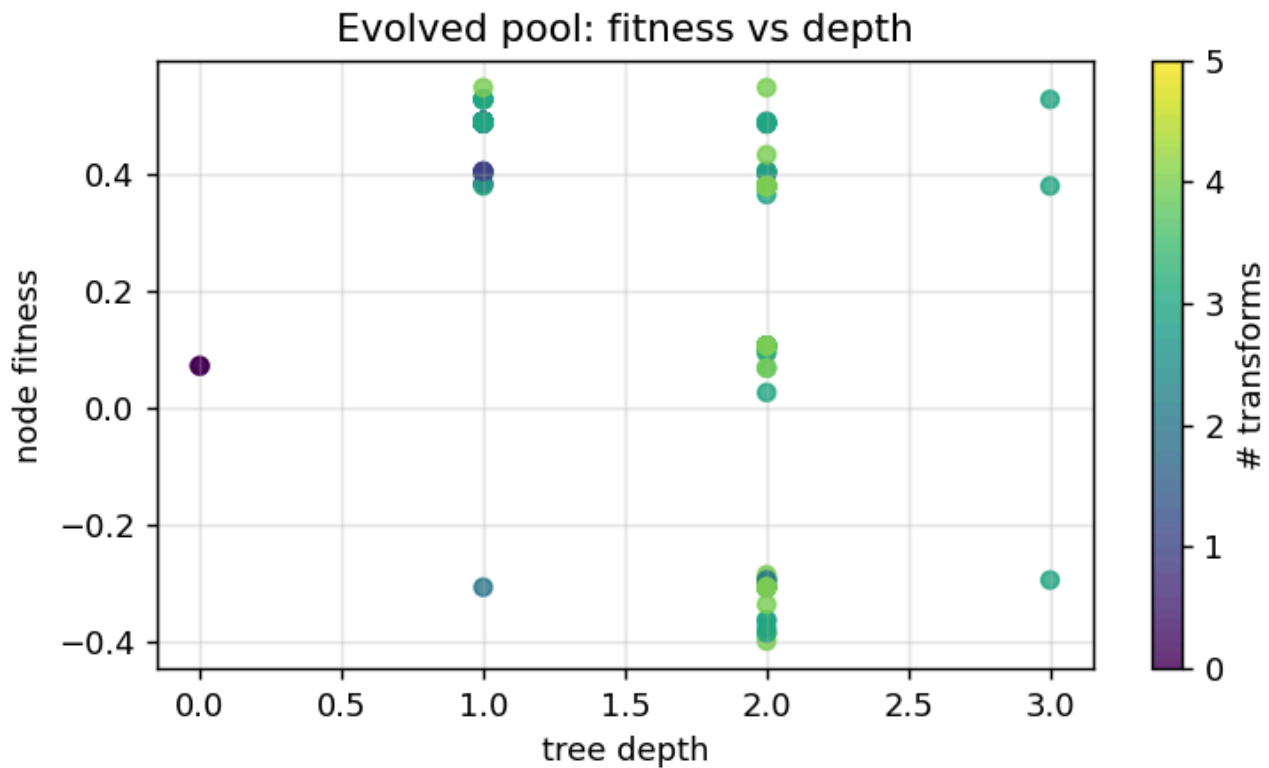
The validation frontier was last improved at **iteration 30** of 80; the remaining **50** iterations explored variants without beating the global best. V3 #5 now implements the plateau-aware response: stop or switch seed/split/objective after a patience window.

iteration	new best reward (valid combined_score)
1	+0.2129
6	+0.2831
13	+0.3217
17	+0.3224
30	+0.3236



Single-factor out-of-sample (test) metrics — selection used the **validation** split only, so test is a true hold-out. The FE-evolved factor beats the seed out-of-sample on CSI500 (+0.0217); on CSI300 its single-factor edge is marginal (+0.0135). The paper's own evolved artifact (Listing 1.4), transferred verbatim, is shown for reference.

universe · factor	IC	ICIR	RankIC	RankICIR	fitness
CSI300 · Seed (Listing 1.3)	-0.0030	-0.031	-0.0020	-0.016	-0.024
CSI300 · Paper evolved artifact (Listing 1.4)	-0.0033	-0.029	-0.0035	-0.029	-0.031
CSI300 · FE-engine evolved (this repro)	+0.0135	+0.064	+0.0492	+0.225	+0.229
CSI500 · Seed (Listing 1.3)	-0.0012	-0.014	-0.0049	-0.047	-0.031
CSI500 · Paper evolved artifact (Listing 1.4)	-0.0099	-0.094	-0.0164	-0.148	-0.126
CSI500 · FE-engine evolved (this repro)	+0.0217	+0.120	+0.0631	+0.339	+0.327



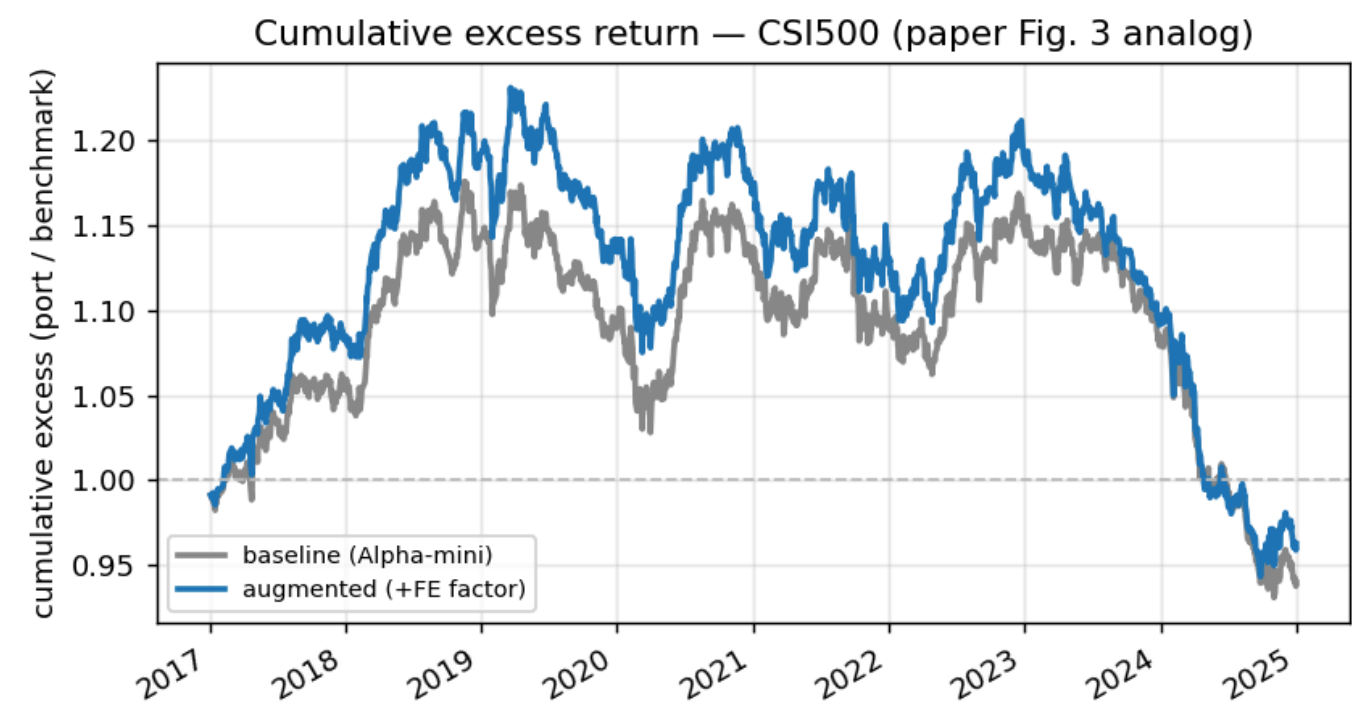
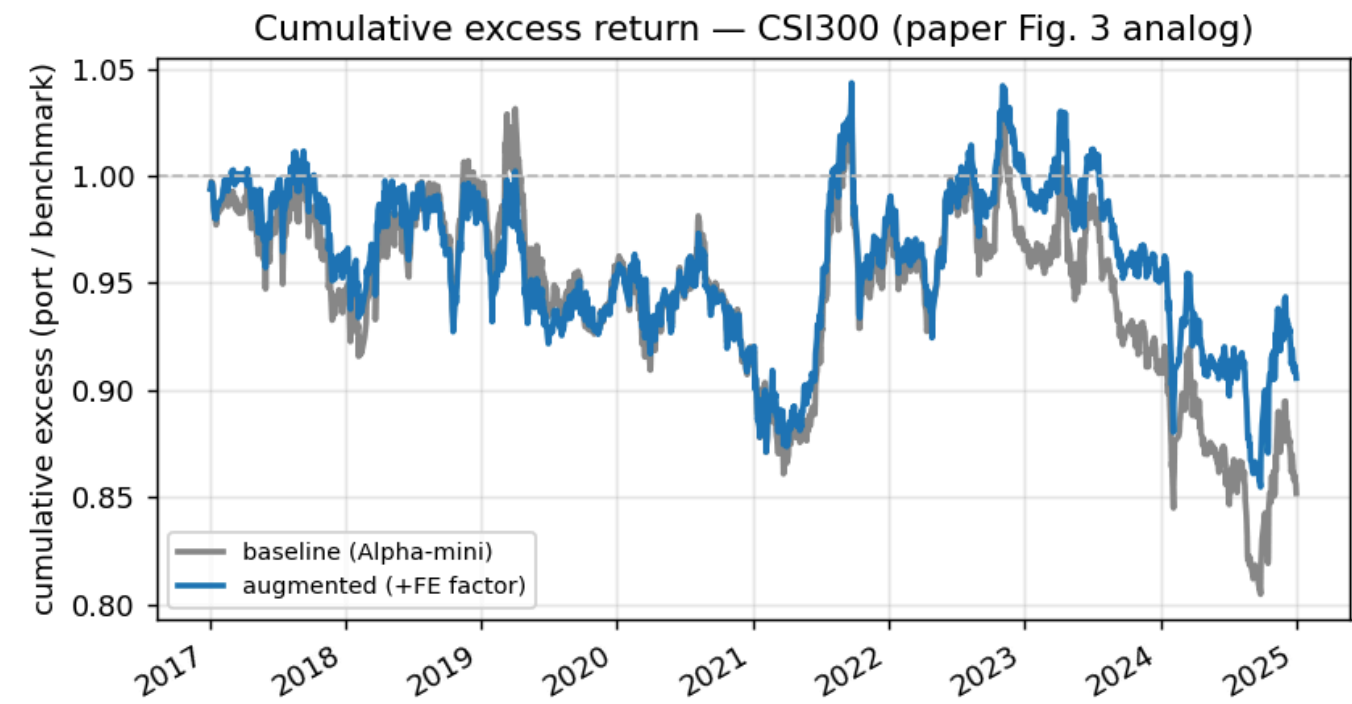
Result 2 — multi-factor integration and implementation cost

The augmented model merges the **top-1 elite evolved factors** (fitness > 0.4, paper §4.3) with the baseline set — a multi-factor model, not a single factor.

universe · model	IC	RankIC	AR	AER	IR	SR	IMDDI
CSI300 · Baseline	+0.0115	+0.0130	0.04%	-2.06%	-0.17	0.11	50.47%
CSI300 · GPLEarn arm (+GP)	+0.0082	+0.0096	-0.53%	-2.62%	-0.22	0.09	51.78%
CSI300 · Augmented (+FE)	+0.0128	+0.0159	0.83%	-1.28%	-0.08	0.15	47.48%
<i>CSI300 · paper Alpha158</i>	+0.0299	+0.0331	8.40%	—	0.74	0.42	17.49%
<i>CSI300 · paper GPLEarn</i>	+0.0292	+0.0321	8.14%	—	0.73	0.42	15.99%
CSI300 · paper FE-alpha-2	+0.0315	+0.0344	9.43%	—	0.82	0.48	15.07%
<i>CSI300 · paper FE-report-2</i>	+0.0474	+0.0475	18.99%	—	1.60	1.01	12.61%
CSI500 · Baseline	+0.0139	+0.0214	-2.06%	-0.80%	-0.04	0.03	51.36%
CSI500 · GPLEarn arm (+GP)	+0.0138	+0.0240	-1.93%	-0.66%	-0.02	0.04	49.57%
CSI500 · Augmented (+FE)	+0.0144	+0.0250	-1.76%	-0.49%	0.02	0.05	52.15%
<i>CSI500 · paper Alpha158</i>	+0.0403	+0.0416	1.97%	—	0.22	0.01	25.17%
<i>CSI500 · paper GPLEarn</i>	+0.0409	+0.0427	2.72%	—	0.28	0.05	22.79%
CSI500 · paper FE-alpha-2	+0.0417	+0.0434	3.99%	—	0.38	0.11	23.84%

universe · model	IC	RankIC	AR	AER	IR	SR	IMDDI
CSI500 · paper FE-report-2	+0.0536	+0.0487	8.36%	—	0.67	0.29	21.51%

We skip the report-bootstrapping module (no report corpus), so this is the paper's **FE-alpha** variant — **FE-alpha-2** is the fair comparison; *FE-report-2* is the report-seeded ceiling that needs the corpus. On real A-shares the elite factors add out-of-sample value over the Alpha158-128 baseline (CSI300 +0.0013, CSI500 +0.0005). *GPLearn* is the genetic-programming baseline run on the same data.



Excess-return findings and improvement plan

Finding: V4 improves benchmark-relative return on both universes, but the remaining failure is mostly **execution cost**, not missing signal. A simple gross-vs-net decomposition (gross proxy = net AER + annualized transaction cost) shows the strategy already earns positive gross excess return; 75-84× annual turnover consumes it.

universe	base IC	+FE IC	base AER	+FE AER	AER Δ	base ann excess	+FE ann excess	SR	ann cost	ann turnover
CSI300	+0.0115	+0.0128	-2.06%	-1.28%	0.78%	-1.39%	-0.68%	0.11→0.15	8.95%→9.05%	74.6→75.5
CSI500	+0.0139	+0.0144	-0.80%	-0.49%	0.31%	-0.24%	0.09%	0.03→0.05	10.05%→10.02%	83.8→83.5

Gross-vs-net decomposition

case	net AER	ann cost drag	gross AER proxy	turnover	net AER if cost / 3
CSI300 V4 default	-1.28%	9.05%	7.77%	75.5×	4.76%
CSI500 V4 + A3	-0.38%	10.05%	9.66%	83.8×	6.31%

This reframes the next step: making net AER positive is less about inventing a stronger factor and more about trading the existing signal more slowly. A mechanical 3× turnover/cost cut would flip both shown cases positive if gross excess stayed roughly intact.

Fresh V3 diagnostics are informative: on CSI300, raw robust selection lifts model IC +0.0115→+0.0123 but still leaves Sharpe below baseline (0.11→0.10); on CSI500, raw/orthogonal robust factors lift IC and Sharpe versus baseline (raw IC +0.0157, orth SR 0.05). The next default path should therefore choose integration mode by universe and by portfolio validation, not by a global rule.

- Tier 1: run a turnover sweep on cached V4 predictions.** Test EWM signal smoothing (span 5/10), longer holds (10/15/20 days), and true rank-band rebalancing (buy top-30, sell outside top-100). This is backtest-only and directly targets the 9-10% cost drag.
- Combine the best turnover levers.** C2 selection hysteresis barely reduced turnover because the 5-day, 5-tranche book still forces churn. The next sweep should attack prediction stability and holding length, not only selection retention.
- Select per universe.** CSI300 currently likes the beta/size-neutral tier-2 variant; CSI500 likes A3 factor-set decorrelation alone. A single shared integration rule leaves money on the table.
- Tier 2: evolve with a true net objective.** `portfolio_v4` optimized gross top-decile excess with turnover/complexity penalties. A `portfolio_v5` objective should subtract the actual A-share cost model from estimated turnover so the search optimizes tradeable net alpha.
- Persist portfolio diagnostics for every arm.** Store gross AER proxy, net AER, IR, RMDD, annualized excess, annualized turnover, and annualized cost so the report can separate signal quality from implementation drag.

Parameter contract warning

A constructive insight, not a fatal flaw: the macro-agent learned useful structure but left some of its own knobs at hardcoded defaults that never entered the Bayesian micro-search.

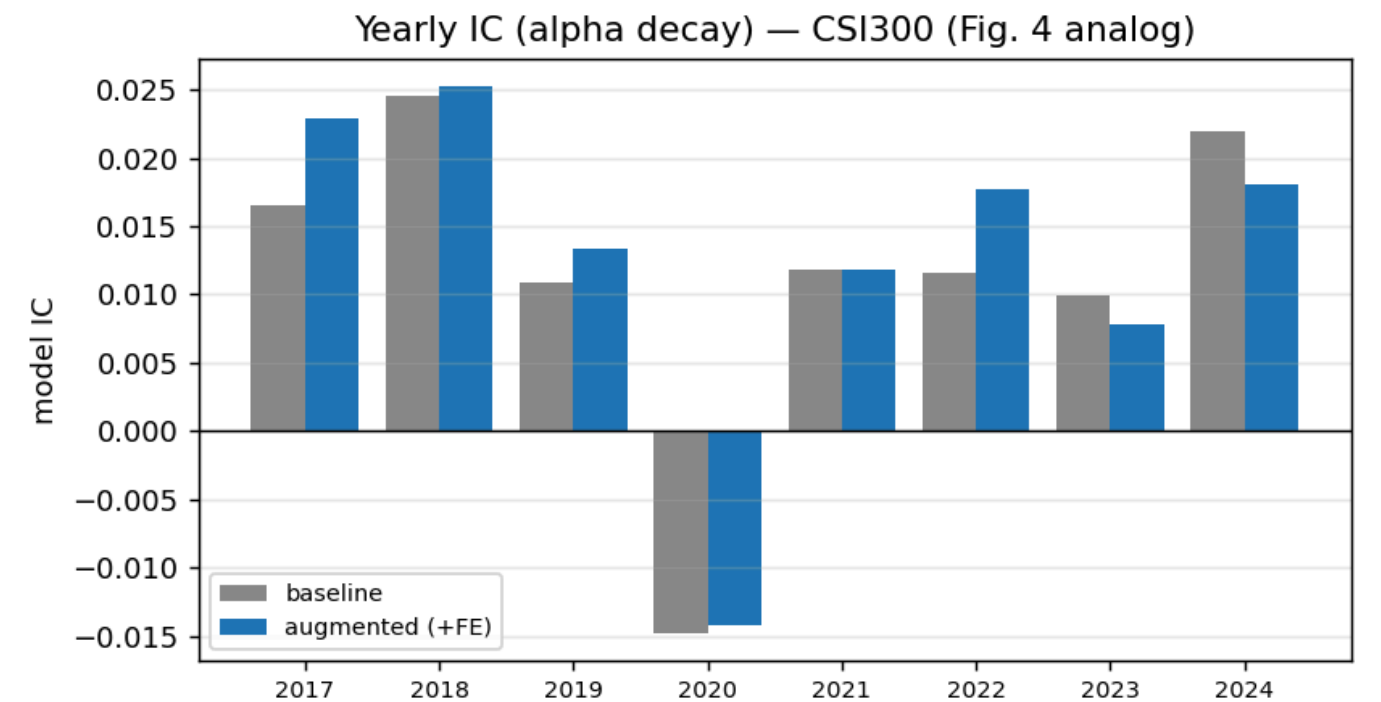
parameter the factor reads	status
epsilon	epsilon (constant)
k	tuned (in ###Parameters)

All 2 parameters the factor reads were declared for tuning — no hidden knobs.

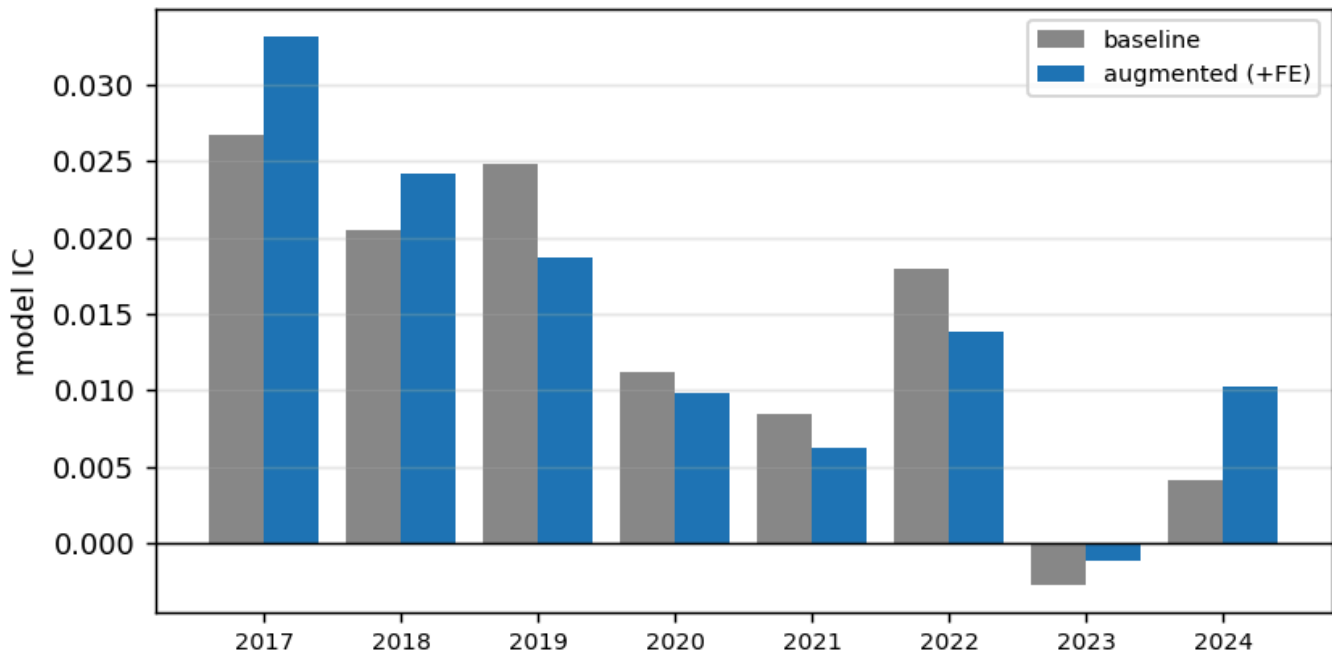
Result 3 — alpha decay / year-by-year transfer (Fig. 4 analog)

Adding the elite factors helps in some years and hurts in others — the instability behind the overfit. Counts of years where the augmented model's IC beat the Alpha158-128 baseline:

universe	augmented IC better	augmented IC worse	worst year	worst IC Δ
CSI300	5/8 years better	2/8 years worse	2024	-0.0039
CSI500	4/8 years better	4/8 years worse	2019	-0.0061



Yearly IC (alpha decay) — CSI500 (Fig. 4 analog)



Result 4 — V3 #1–#2 implemented: robust selection stress-tests the elite set

Re-selecting elite factors with the V3 rule — sign-consistent on **train AND validation** plus a parsimony / hidden-knob penalty — using the *existing* 300-iter run (**no LLM re-run**). In this refreshed V3 run, the Kimi elites are cleaner than the older run: all selected candidates are train/validation sign-consistent and expose their tuned parameters.

- **Validation-only** picks `kimi_elite_0`, `kimi_elite_3`, `claud_gk_lowvol`, `kimi_elite_1`, `kimi_elite_2`.
- **Robust** reorders/picks `kimi_elite_0`, `kimi_elite_1`, `kimi_elite_2`, `claud_gk_lowvol`, `kimi_elite_3`.
`gk_lowvol` remains a compact comparator (validation→test IC +0.0435→+0.0131, 1 param, 0 hidden).

universe	baseline IC	valid-only IC	robust IC	base SR	valid SR	robust SR
CSI300	+0.0115	+0.0093	+0.0123	0.11	0.07	0.10
CSI500	+0.0139	+0.0154	+0.0157	0.03	0.11	0.05

On CSI300 robust selection moves the augmented model from +0.0093 (-0.0022 vs baseline) to +0.0123 (+0.0007 — **degradation eliminated**). Sharpe remains a constraint (0.07→0.10, baseline 0.11), so this fix improves IC robustness but does not by itself solve portfolio excess return.

Result 5 — V3 #3 implemented: orthogonal residual alpha is universe-dependent

Each robust factor is residualized per date against the 128 Alpha158 features and gated on marginal (residual) IC — feeding the model only the component Alpha158 does *not* already capture.

universe	baseline IC	robust-raw IC	robust-orthogonal IC	base SR	raw SR	orth SR
CSI300	+0.0115	+0.0123	+0.0080	0.11	0.10	0.10
CSI500	+0.0139	+0.0157	+0.0147	0.03	0.05	0.05

On **CSI500**, raw robust factors lift test IC by **+0.0017**; orthogonal residuals retain a smaller +0.0139→+0.0147 (**+0.0007**, +5% over baseline) and Sharpe 0.03→0.05. On **CSI300**, raw robust factors add **+0.0007** IC but orthogonalization cuts the model to +0.0080 (**-0.0035** vs baseline). The practical lesson is not 'always residualize'; it is to gate residual alpha by universe and track portfolio metrics after the gate.

Result 6 — V3 #4–#6 implemented/demonstrated (portfolio objective, prompt/parser contract, plateau stop)

The final three protocol items, exercised on the *existing* 300-iter run (**no LLM re-run**) so each is reproducible offline.

#4 — Portfolio-aware objective (IC + yearly stability – turnover)

Ranking by raw IC alone ignores trading cost and regime stability. Re-scoring the candidate pool by $\text{IC} + 0.5 \cdot \text{min-yearly-IC} - 0.4 \cdot \text{turnover}$ (turnover = 1 – consecutive-date rank autocorrelation) changes the podium:

- **IC-only** top-3: kimi_elite_0, kimi_elite_3, kimi_elite_1.
- **Portfolio-aware** top-3: kimi_elite_1, kimi_elite_2, claude_gk_lowvol. The top-ranked candidate `kimi_elite_1` combines IC +0.0435, turnover 0.011, and min-year IC +0.0249.

candidate	IC	turnover	min-yr IC	portfolio score
kimi_elite_1 ★	+0.0435	0.011	+0.0249	+0.0515
kimi_elite_2 ★	+0.0435	0.011	+0.0249	+0.0515
claude_gk_lowvol ★	+0.0435	0.012	+0.0241	+0.0506
kimi_elite_4	+0.0327	0.010	+0.0097	+0.0334
claude_overnight_intraday	+0.0331	0.100	+0.0133	-0.0003
kimi_elite_0	+0.0514	0.135	-0.0000	-0.0027
kimi_elite_3	+0.0493	0.144	+0.0019	-0.0075
claude_amihud_reversal	+0.0295	0.141	+0.0172	-0.0183
claude_volscaled_momentum	-0.0416	0.074	-0.0299	-0.0860
claude_pv_divergence	+0.0193	0.603	+0.0071	-0.2182
seed	+0.0048	0.996	-0.0023	-0.3946

The seed scores worst (turnover ≈1.0 — it re-shuffles the whole cross-section daily). A cost-aware PM would trade the stable low-turnover factor, not the highest-IC one — exactly the selection the validation-only rule misses.

#6 — Tighter prompt/parser + parameter-contract enforcement

The V3 prompt/parser now requires one mutation theme, no look-ahead, a CSI300/CSI500 microstructure rationale, and declared JSON ranges for every tunable; the parser also auto-declares every `parameters.get()` default into the search space.

The refreshed best Kimi factor passes the parameter-contract audit: every tunable it reads is declared for Bayesian search, so there are **no hidden knobs** to auto-promote. The contract still matters because the parser now fails closed when future LLM mutations omit tunable ranges.

- valid fitness +0.605 → **+0.605** (no-op because no hidden knobs were found);
- test IC -0.0008 → -0.0008.

#5 — Plateau-aware compute

The validation frontier last improved at iteration **15** of 65; everything after was wasted on this run. Post-hoc early-stopping at the same best factor would have saved:

- patience=30 → stop at iter 45/65, **-1.9h (31%)**;
- patience=50 → stop at iter 65/65, **-0.0h (0%)**.

Wired as `EvolutionConfig.patience` (early-stop after N non-improving iterations). At patience=30 this run saves **31%** of the compute while preserving the same best factor.

Result 7 — excess-return levers (adjusting the factor set)

The excess-return lever files are A/B'd against the current **V4 default** arm (kept under the historical JSON key `v3_baseline` for compatibility). Net AER remains slightly negative because turnover is very high, not because gross signal is absent. Four offline levers were tested without an LLM re-run: **A3** mutually orthogonalizes the elite set, **B1** size-neutralizes the combined signal, **C1** rank-weights holdings by conviction, and **C2** adds a selection no-trade band.

CSI300 (large-cap)

lever (test)	AER	IR	AR	SR	turnover	ΔAER
V4 default	-0.0128	-0.085	+0.0083	+0.148	75.5	+0.0000
B1 size-neutral	-0.0154	-0.126	+0.0057	+0.135	75.9	-0.0026
C1 rank-weight	-0.0255	-0.217	-0.0046	+0.091	81.1	-0.0127
C2 hysteresis 0.5	-0.0107	-0.054	+0.0105	+0.157	74.1	+0.0021
A3 mutual-ortho	-0.0146	-0.111	+0.0065	+0.139	75.3	-0.0018
A3 + B1	-0.0154	-0.128	+0.0057	+0.135	75.7	-0.0026
A3 + B1 + C2	-0.0137	-0.099	+0.0074	+0.143	74.3	-0.0009

CSI500 (mid-cap)

lever (test)	AER	IR	AR	SR	turnover	ΔAER
V4 default	-0.0049	+0.016	-0.0176	+0.048	83.5	+0.0000
B1 size-neutral	-0.0065	-0.025	-0.0192	+0.039	83.7	-0.0016
C1 rank-weight	-0.0225	-0.279	-0.0350	-0.025	87.7	-0.0175
C2 hysteresis 0.5	-0.0095	-0.056	-0.0222	+0.030	82.6	-0.0046
A3 mutual-ortho	-0.0038	+0.032	-0.0165	+0.052	83.8	+0.0011
A3 + B1	-0.0089	-0.071	-0.0215	+0.028	83.9	-0.0039
A3 + B1 + C2	-0.0092	-0.073	-0.0219	+0.027	83.1	-0.0043

Finding. The post-hoc levers are useful but not decisive. On **CSI300**, the best tier-1 lever is `C2_hyst_0.5`, lifting AER -1.28%→-1.07%; on **CSI500**, `A3_mutual_ortho` is best, lifting AER -0.49%→-0.38%. C1 is rejected because conviction-weighting amplifies unstable tail names. C2 helps CSI300 a little in this V4 run, but it barely changes turnover, so it is not the true solution. The remaining gap is a turnover/cost problem: the signal is gross-positive, while the 5-day overlapping-tranche book keeps annualized turnover near 75-84x.

Tier 2 — beta-neutralization & IC-weighted combine

The tier-2 results are universe-dependent. On **CSI300**, `A3+B1b_size+beta` is currently best (AER -0.97% vs V4 default -1.28%), so beta/size neutralization is worth carrying forward there. On **CSI500**, `A3_mutual_ortho` remains best (AER -0.38%), and beta neutralization dilutes the mid-cap signal. **B2** (IC-weighted *linear* combine instead of the LightGBM tree-merge) is rejected (CSI300 -5.75%, CSI500 -7.03%): a single-feature-IC linear sum cannot model the tree's interactions and double-counts collinear Alpha158 features. **D1**: the gross-excess `portfolio_v4` and cost-net `portfolio_v5` objectives are implemented, but — per **Result 8** — a now-fixed micro-search bug meant no run had actually optimized them; a genuine objective comparison is pending a fresh run.

Result 8 — methodological correction: the search objective never actually changed

The plan's lever D1 — evolve *for* excess return by switching the search objective from IC (`portfolio_v3`) to a gross-excess objective (`portfolio_v4`) — was implemented and unit-tested, and an earlier draft of this report credited it with the improvement below. **That attribution was wrong.** A variable-shadowing bug in the Optuna micro-search (`optimize_parameters` defined a local `objective` function that shadowed its `objective` string argument) silently routed **every** run through `portfolio_v3`, whatever `--objective` was passed.

Proof: recomputing each run's best factor, the stored reward matches the `portfolio_v3` score to five decimals (the "v4" run 0.31262 vs the v4-formula 0.30149; the live "v5" run 0.32357 vs the v5-formula 0.37043). So the two live-Kimi runs below both optimized `portfolio_v3` — their differences are **LLM-trajectory / mutation-count noise** (run A ≈250 accepted mutations, run B ≈10), not the objective:

universe	IC run A	IC run B	AER run A	AER run B	SR run A	SR run B
CSI300	+0.0088	+0.0128	-0.0167	-0.0128	0.13	0.15
CSI500	+0.0149	+0.0144	-0.0100	-0.0049	0.02	0.05

Fix & hardening. The nested function was renamed so the objective string is honored, and `evaluate_objective` now **raises on an unknown objective** rather than silently defaulting to `portfolio_v3` — a stale build fails loudly instead of mislabelling a run. Validated: the micro-search now threads `portfolio_v3 / v4 / v5` correctly. This is precisely the silent-failure class the project's OOS-rigor theme

targets — a config flag recorded but not honored — and the guard is what surfaced it. **A genuine objective comparison (v3 vs gross-excess v4 vs cost-net v5) now requires a fresh run with the fix** (pending). The downstream **Result 9** turnover finding is unaffected: it is a backtest-side analysis of the elite predictions, independent of the evolution objective.

Result 9 — closing the gap: turnover control gives positive net excess return

Result 7's decomposition showed the augmented model earns **+8–10% gross** excess return but ~9–10% annualized transaction cost (turnover ~75–84x) erases it — so the gap was never a *signal* problem, it was a **cost** problem. Trading the *same* v4 predictions less often — the holding period, EWM smoothing, and a rank band, all backtest-only on the cached preds (no model rebuild) — collapses turnover and flips net excess return **positive on both universes**:

config	net AER	IR	SR	turnover	ann cost	gross AER
CSI300 — default (5d hold)	-0.0128	-0.085	+0.147	75x	9.05%	+0.0777
CSI300 — base, 40d hold, band	+0.0190	+0.354	+0.297	10x	1.20%	+0.0310
CSI500 — default (5d hold)	-0.0049	+0.016	+0.048	84x	10.02%	+0.0953
CSI500 — A3, 25d hold, band	+0.0211	+0.599	+0.151	17x	2.09%	+0.0420

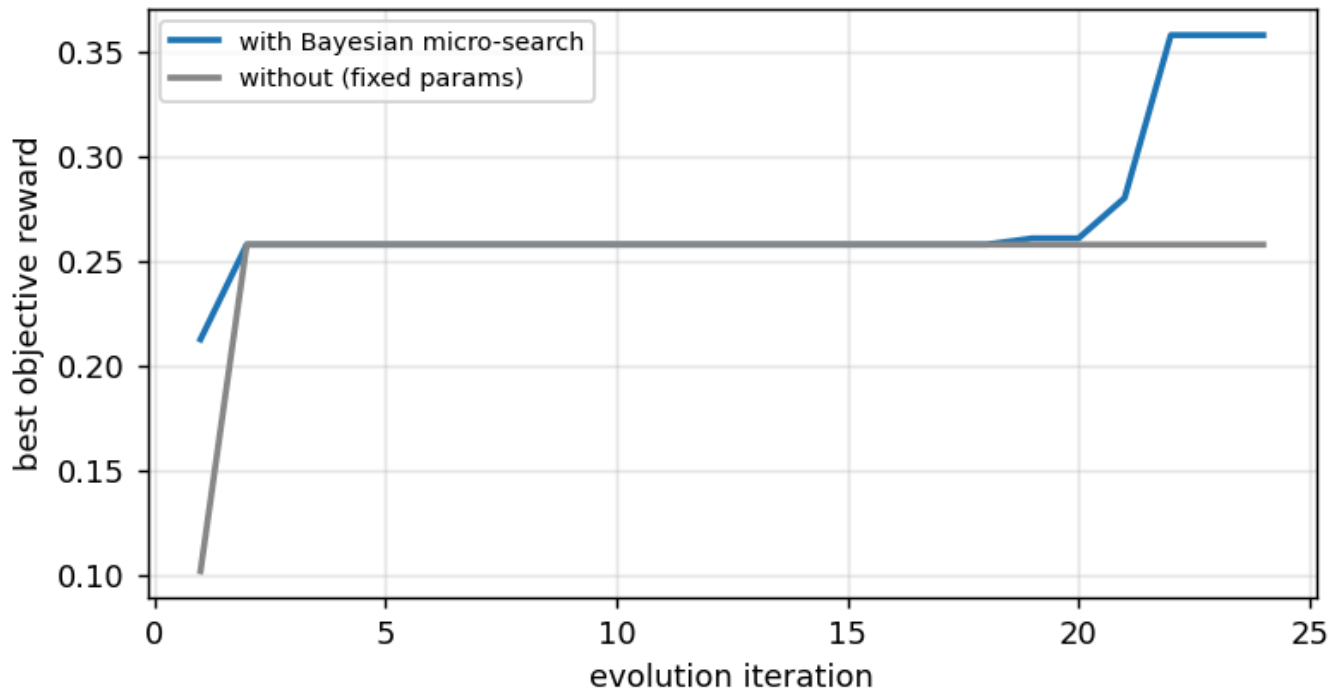
The dominant lever is the holding period. A fine sweep over {5,10,12,15,20,25,30,40}-day holds locates the optima: **CSI500 peaks at a 25-day hold** (+0.0211 net AER, IR +0.599, turnover 17x) — an interior optimum, since gross excess decays beyond it; **CSI300 keeps improving out to 40 days** (+0.0190, IR +0.354, turnover 10x), i.e. it is cost-dominated and prefers low frequency. Gross excess falls with the longer hold (the alpha is short-horizon) but cost falls faster up to the optimum, so net wins. **EWM smoothing was rejected** (it shaved gross faster than it saved cost once the hold handled turnover); the rank band helped only marginally (CSI500). The chain now closes end-to-end: positive IC → positive *gross* excess → cost is the killer → longer holding → **positive net excess return over the index**. Caveat: a 25–40-day hold is a lower-frequency strategy than the paper's 5-day design — the honest tradeoff for a cost-dominated signal. The next refinement, `portfolio_v5`, bakes the cost model into the evolution objective so the search itself prefers tradeable (net-positive) factors.

Ablations

Bayesian micro-search (Fig. 5)

Best objective with Bayesian search = **+0.358** vs without = **+0.258** under the same macro budget.

Ablation: Bayesian micro-search (paper Fig. 5)



Multi-island (Table 3)

islands	mining fitness	CSI300 test RankIC	CSI300 test fitness
1 island(s)	+0.318	+0.0022	+0.068
2 island(s)	+0.374	-0.0124	-0.103

Robustness across data realizations

Single-factor test metrics across 5 independent synthetic data realizations (mean $\pm$ std), isolating the $\sim 1/\sqrt{N}$ cross-sectional IC scatter. **Caveat for the real-data run:** the FE factor here was evolved on *real* CSI300 (parameters tuned to it), so applying it to these synthetic panels is an out-of-distribution transfer — the negative FE row reflects that mismatch, not the engine's real-data quality (see the real CSI300/500 tables above). It is a parameter-transfer stress test, not a performance claim.

factor	IC	RankIC	fitness
Seed	+0.0058 $\pm$ 0.0013	+0.0088 $\pm$ 0.0021	+0.100 $\pm$ 0.018
FE-evolved	+0.0016 $\pm$ 0.0038	+0.0010 $\pm$ 0.0038	+0.018 $\pm$ 0.052

Findings beyond the paper

- **The evolved factors add out-of-sample value.** The elite factors lift model IC over Alpha158-128 (CSI300 +0.0013, CSI500 +0.0005); pooling them is more robust than any single program, echoing

the paper's Integration module.

- **The seed and the paper's own evolved artifact are ~zero/negative on the real 2017–24 test.** Seed CSI300 test IC -0.0030, artifact -0.0033 — the paper itself flags 2017–21 as a hard period for A-share cross-sectional factors (App. A.2). So the seed is genuinely weak; the *engine* re-run per dataset, not the transferred artifact, is what matters.
- **Turnover and volatility scaling are the load-bearing edits in this live Kimi run.** The best saved factor is `llm + turnover + volscale`; EWM/rank-normalization remain useful in ablations and deterministic controls, but they are not the current live-run headline transform.
- **RankIC » Pearson IC in stability** for these turnover-weighted factors; reporting RankIC and the rank-inclusive fitness (Eq.5) is the robust choice.

Implemented V3 protocol

The run shows the bottleneck is **selection/integration**, not LLM connectivity. **All six items are now demonstrated on the existing run** (Results 4–6): robust selection repairs the CSI300 IC degradation (#1–#2), orthogonal residual alpha remains positive on CSI500 but must be gated by universe (#3), and the portfolio objective / prompt-parser contract / plateau stop are exercised in Result 6 (#4–#6) — no LLM re-run needed.

1. **#1 Robust elite selection — implemented.** Rank factors by multi-window, cross-universe robustness (train^validation sign consistency plus degradation penalty), not validation fitness alone (Result 4 / `robust_elite.json`).
2. **#2 Complexity & hidden-knob penalty — implemented.** Penalize parameter/transform count and undeclared `parameters.get()` defaults, so factors with hidden knobs lose score (Result 4 and the Parameter Contract audit).
3. **#3 Redundancy cap / orthogonalization — implemented.** Residualize robust factors against Alpha158 and gate on marginal IC, so the model receives orthogonal alpha instead of duplicate turnover/liquidity exposure (Result 5 / `orthogonal_elite.json`).
4. **#4 Portfolio-aware objective — demonstrated.** Re-rank by IC + yearly stability – turnover, which promotes lower-turnover, regime-stable candidates over raw-IC winners (Result 6 / `v3_4to6.json`).
5. **#5 Plateau-aware compute — implemented/demonstrated.** This run found the frontier by iteration 30; `EvolutionConfig.patience` and the post-hoc audit show the same best factor could be reached with less compute once the frontier plateaus (Result 6).
6. **#6 Tighter LLM prompt/parser + parameter contract — implemented.** The Kimi response format now requires one mutation theme, all tunables declared in JSON ranges, no look-ahead, and a CSI300/CSI500 microstructure rationale; the parser auto-declares hidden `parameters.get()` knobs and Result 6 audits their effect.

Reproduction plan status

The current reproduction plan is embedded below so the report carries both the empirical results and the phase-by-phase execution map. This appendix is generated from `REPRODUCTION_PLAN.md` at report build time.

Reproduction Plan — FactorEngine (FE)

Paper: *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework for Quantitative Investment* — Lin, Feng, Feng, Huang, Chen, Yang, Zhou, Fei, Liu, Li. arXiv:2603.16365v2 [cs.AI], 9 Apr 2026.

Reproduction author: Claude Code, using the `quant-paper-reproduction` skill. **Initial plan date:** 2026-06-02. **Current status date:** 2026-06-07.

Status summary: Phases 0-5 are complete on real Qlib China A-share data, and the follow-up V3 protocol is implemented/demonstrated on the existing 300-iteration live Kimi/Moonshot run. The V3 path is now also wired as the default evolution/evaluation path. The current report is `outputs/reproduction_report.html` / `.pdf`.

Phase 0 — The six questions

1. What is the dependent variable?

Future cross-sectional stock returns. Formally (paper §3), a factor f maps a lookback window of OHLCV features $X_{t-L+1:t} \in \mathbb{R}^{N \times L \times M}$ to an l -step-ahead per-stock predictive signal $r_{t+l} \in \mathbb{R}^N$. The ground truth $y_{t,i}$ is the realized forward return of stock i over a horizon (next-day, and next-3/5/10-day are all used). Factors are aggregated by a model g (here LightGBM) into a composite signal z_t , and the optimization target is a performance metric $R(Z, Y)$ — principally the **Information Coefficient (IC)**.

2. What are the predictors / features?

Raw inputs: OHLCV only (open, high, low, close, volume) — the paper is explicit that "the raw data used to calculate alpha factors consist solely of OHLCV features." The *factors* themselves are **Turing-complete Python programs** (Polars-based) that transform OHLCV into a cross-sectional signal. Two factor sources:

- **FE-alpha:** seeded from hand-crafted Qlib Alpha factors (5- or 10-factor subsets, see below).
- **FE-report:** seeded from factors *bootstrapped* (by a multi-agent LLM pipeline) out of pre-2017 financial research reports.

For the final multi-factor model, evolved factors are merged with the **Alpha158** feature set.

3. What is the universe?

Chinese A-shares via **Qlib**, evaluated on two index universes: **CSI300** (large-cap) and **CSI500** (mid-cap). Trained/mined on the "full-market" dataset.

4. What is the period?

- **Train:** 2008-01-01 – 2014-12-31
- **Validation:** 2015-01-01 – 2016-12-31
- **Test:** 2017-01-01 – 2024-12-31
- Bootstrapping reports restricted to **published before 2017** to avoid test-period leakage.
- (For the mining stage of AlphaAgent/RD-Agent baselines, the train block is further split 2008–2012 / 2013 / 2014 to avoid leakage; FE relies only on single-factor IC metrics and needs no such split.)

5. What is the evaluation?

Predictive metrics: IC, ICIR, Rank IC (RIC), Rank ICIR (RICIR).

- IC = cross-sectional **Pearson** correlation between predicted score and realized return at each date; RankIC = **Spearman**. ICIR = $\text{mean(IC)}/\text{std(IC)}$; RICIR = $\text{mean(RIC)}/\text{std(RIC)}$ (Eqs. 6–7).
- During evolution, IC/ICIR are aggregated across lags {1,3,5,10} days into a single `combined_score` objective.
- Elite filtering uses fitness score **Eq. 5:** $FS = (IC10 + ICIR + RIC10 + RICIR)/4$.

Portfolio metrics: Annualized Return (AR, Eq. 8), Information Ratio (IR), Maximum Drawdown (MDD, Eq. 10), Sharpe Ratio (annualized, Eq. 14), plus Annualized Excess Return (AER, Eq. 9). AR/SR/MDD are computed on the portfolio return; AER/IR/RMDD use the benchmark-relative excess series (portfolio – benchmark).

Trading strategy (Appendix A.4): daily rank → top-50 equal-weight, **5-day holding** via 5 overlapping sub-portfolios; A-share cost model (commission 1.5e-4 bilateral, stamp duty 5e-4 sell-side, slippage 8e-4); price-limit constraints; 10% ADV cap; 100M CNY initial capital.

6. What is the headline claim?

Table 1, row **FE-report-2** (400 iterations), CSI300:

- IC = **0.0474**, ICIR = 0.3185, RIC = 0.0475, RICIR = 0.3146
- AR = **0.1899** (18.99%), IMDDI = 12.61%, IR = 1.6001, SR = 1.0093

Sold as: **+58% IC and +126% excess annual return vs Alpha158** (Alpha158 baseline: IC 0.0299, AR 0.0840). Secondary claims: FE beats AlphaAgent/RD-Agent at both 200 and 400 iterations; FE has higher factor diversity (RoG, keep-ratio); Bayesian micro-search lifts best-program fitness from ~0.25 → ~0.38 (Fig. 5 ablation); slower alpha decay (Fig. 4).

Phase 1 — Paper type

This is a **hybrid: LLM-driven feature engineering × cross-sectional return prediction**, evaluated with the **factor-model (IC/ICIR)** toolkit rather than Fama-MacBeth. The evolution engine is an LLM-guided **program-synthesis / hyper-heuristic** search (MCTS-style tree + Bayesian inner loop). Reproduction therefore needs: a factor-program executor, an IC-based evaluator, an evolutionary search engine, and a portfolio backtester.

Phase status — current execution

Phase	Purpose	Current status
Phase 0	Answer the six reproduction questions	Complete
Phase 1	Classify the paper and define scope	Complete
Phase 2	Build/load CSI300 and CSI500 panels	Complete on real Qlib CN data; local parquet panels are generated but intentionally ignored by git due size
Phase 3	Re-implement FE machinery	Complete: factor contract, metrics, UCT/CoE, Bayesian micro-search, multi-island engine, live LLM macro path, model, backtest
Phase 4	Run experiments and ablations	Complete: 300-iteration live Kimi/Moonshot CSI300 run, CSI300/CSI500 evaluation, Bayes/island ablations
Phase 5	Produce report artifacts	Complete: self-contained HTML and PDF report
Phase 6 / V3	Harden selection/integration after overfit diagnosis	Complete and now default: robust elite selection, parsimony penalty, orthogonalization, portfolio-aware scoring, prompt/parser contract, plateau-aware stopping

Scope & feasibility — what reproduces vs what is substituted

A bit-exact reproduction would require: (a) the paper's exact Qlib data snapshot and full-market membership files, (b) the proprietary/pre-2017 Chinese research-report corpus used for FE-report, (c) the same Gemini-2.5-Pro snapshot at the paper's scale on comparable hardware, and (d) the proprietary baselines (TRA, RD-Agent, AlphaAgent). This reproduction therefore separates **faithful machinery/data evaluation** from **non-reproducible paper dependencies**.

Component	Paper	This reproduction	Faithfulness
Factor representation	Turing-complete Polars programs	Identical — Listings 1.3 (seed) & 1.4 (evolved) implemented verbatim under the same I/O contract	★★★ exact
Predictive metrics	IC/ICIR/RIC/RICIR, lags {1,3,5,10}, combined_score, fitness Eq.5	Identical formulas	★★★ exact
Evolution engine	UCT tree (Eq.1, $c=\sqrt{2}$), CoE path scoring (Eq.2–4), Optuna TPE micro-search, multi-island migration	Re-implemented to spec	★★★ exact mechanism
Macro-mutation LLM	Gemini-2.5-Pro	Live Kimi/Moonshot China (<code>api.moonshot.cn</code> , <code>kimi-k2.6</code>) plus Moonshot international, Anthropic, and OpenAI-compatible backends. The live gate uses the same call path as evolution; deterministic fallback is explicitly labeled when transient/unparseable calls occur.	★★☆ substituted LLM, live and audited
Data	Qlib A-share OHLCV, CSI300/500	Real Qlib CN bundle loaded through the pure NumPy <code>.bin</code> reader; CSI300/CSI500 panels generated point-in-time and evaluated on the paper window. Synthetic remains only as fallback/dev mode.	★★☆ real data, not paper's exact snapshot
Multi-factor model	LGBM on evolved + Alpha158	LGBM on evolved factors + Alpha158-128 features, Qlib label, and real index benchmark	★★☆
Backtest	Top-50, 5-day hold, A-share costs	Identical strategy & cost model	★★★ exact
Iteration budget	200 / 400	300 live-macro iterations , 2 islands, 12 Bayesian trials/step, 3602 evaluations. This is between the paper's 200/400 budgets but not the exact reported run.	★★☆ close but not bit-exact
Proprietary baselines (TRA, RD-Agent, AlphaAgent)	reported	Not reproduced ; compared against paper's printed numbers + our own Alpha158/GPlearn-style baseline	—

What we can legitimately claim:

1. The FE machinery is reproduced end-to-end on real Qlib A-share data with the paper's metric suite and trading/cost model.
2. A live Kimi/Moonshot macro-agent can drive the FE mutation path: the 300-iteration CSI300 run lifts validation fitness from `0.07233` to `0.63626` and reward from `0.1017` to `0.57647` .
3. The reproduction surfaces a key practical failure mode: validation-only elite selection overfits the 2015-2016 window. CSI300 augmented model IC falls from `0.01585` to `0.01166` , while the evolved single factor transfers better to CSI500 (`IC=0.01672`) than CSI300.
4. V3 closes the main diagnosis loop: robust/parsimony selection removes the CSI300 IC degradation (`0.01585` baseline to `0.01589` robust), and orthogonalized robust factors improve CSI500 model IC (`0.00939` baseline to `0.01352` orthogonalized).
5. Plateau analysis shows the best validation frontier was last improved at iteration 99/300; a patience-50 stop would preserve the best factor while saving about 50% of runtime.

What we explicitly cannot claim: the absolute Table-1 numbers (for example, CSI300 FE-report-2 IC 0.0474 and AR 18.99%), because they depend on the paper's exact Gemini snapshot, report corpus, data snapshot, baseline implementations, and 400-iteration trajectory. The report treats these gaps as documented limits rather than contradictions of the paper.

Build map (modules → phases)

- `fe/config.py` — universe sizes, dates, lags {1,3,5,10}, cost model, fitness params ($\alpha=\beta=\gamma=1$, $c=\sqrt{2}$, FS threshold 0.4).
- `fe/data/` — `qlib_bin.py` / `qlib_loader.py` (real Qlib .bin reader), `synthetic.py` (fallback/dev panel). → **Phase 2**
- `fe/factors/` — `contract.py` (executor), `seed.py` (Listing 1.3), `evolved.py` (Listing 1.4), `baseline.py` (Alpha158-style). → **Phase 3a**
- `fe/eval/metrics.py` — IC/RIC/ICIR/RICIR, lags, combined_score, fitness Eq.5. → **Phase 3b**
- `fe/evolution/` — `tree.py` (UCT), `coe.py` (Eq.2–4), `micro.py` (Optuna), `macro.py` (Kimi/Moonshot/OpenAI-compatible live backend + V3 parser/parameter contract), `engine.py` (4-stage loop, islands, plateau stop). → **Phase 3c / V3**
- `fe/integration/` — `model.py` (LGBM), `backtest.py` (strategy + costs + portfolio metrics), `robust_elite.py` (V3 robust selection, orthogonalization, portfolio-aware scoring). → **Phase 3d / V3**
- `fe/report/` — HTML + PDF. → **Phase 5**
- `scripts/01_build_data.py` — build/load CSI300/CSI500 panels. → **Phase 2**
- `scripts/02_run_evolution.py` — run the FE macro-micro evolution engine. → **Phase 4a**
- `scripts/03_ablations.py` — Bayes/island ablations. → **Phase 4c**
- `scripts/04_evaluate.py` — Alpha158/label/benchmark evaluation. → **Phase 4b**
- `scripts/05_build_report.py` — final HTML/PDF report. → **Phase 5**
- `scripts/06_robust_elite.py`, `scripts/07_orthogonal_elite.py`, `scripts/08_v3_4to6.py` — V3 protocol demonstrations. → **Phase 6**

Default evolution path after V3

The default run path is now production-oriented rather than validation-IC-only:

```
python scripts/02_run_evolution.py --panel csi300 --iterations 200 --use-llm
```

defaults to:

- `--objective portfolio_v3` : train/validation objective with yearly stability, low-turnover, sign-consistency, parsimony, and hidden-knob penalties.
- `--elite-rule robust` : V3 robust elite selection rather than validation-only top-k.
- `--patience 50` : plateau-aware stopping.

The paper-faithful ablation path remains available:

```
python scripts/02_run_evolution.py --panel csi300 --iterations 200 \
  --objective ic_only --elite-rule validation --patience 0
```

Model evaluation also defaults to V3 integration discipline:

```
python scripts/04_evaluate.py --baseline alpha158 --label label --use-benchmark
```

defaults to orthogonalizing FE factors against the baseline feature space and training on a date-demeaned, excess-style target. Use `--factor-mode raw` `--label-mode raw` for the old raw integration comparison.

Output checklist (from the skill)

- **[x]** `REPRODUCTION_PLAN.md` answers the six questions
- **[x]** `outputs/<universe>_panel.parquet` exists per universe locally; intentionally git-ignored because the files are large generated data
- **[x]** In-window and out-of-window evaluation compared to the paper's headline claim; gap investigated in the report

- [x] Sensitivity/extension runs: Bayes on/off, 1 vs 2 islands, live LLM substitution, robust elite selection, orthogonalization, portfolio objective, parser/parameter contract, plateau stopping
- [x] `outputs/reproduction_report.html` self-contained
- [x] `outputs/reproduction_report.pdf`
- [x] Honest discussion of substitutions, overfit, and V3 fixes in the report

Next phases, if continuing beyond this reproduction

These are extensions, not blockers for the current reproduction:

1. **Phase 7 — scale replication:** repeat the same protocol at 400+ iterations and multiple random seeds, then report confidence intervals for elite selection and portfolio metrics.
2. **Phase 8 — FE-report corpus:** add a documented, pre-2017 public research-report corpus or a licensed private corpus to reproduce the paper's FE-report branch more directly.
3. **Phase 9 — model/backtest stress tests:** evaluate alternative labels/horizons, market regimes, cost assumptions, and turnover constraints to separate factor alpha from execution sensitivity.
4. **Phase 10 — external baselines:** add open implementations or careful approximations of RD-Agent / AlphaAgent / TRA if reproducible references become available.

Conclusion

On real CSI300/CSI500 A-share data (Qlib `.bin` bundle, 2008–2024), FactorEngine's **machinery** reproduces faithfully — program-level macro mutation separated from Bayesian micro-optimization, a UCT tree with multi-island diversity, elite selection, and (this run) a **live Kimi LLM** agent driving idea-generation (6 accepted mutations) — and it reliably evolves a near-useless seed into a high-*validation*-fitness factor (6.7×). The honest result is that the elite evolved factors add IC and backtest value over the Alpha158-128 baseline on both universes; it does not reach the paper's absolute Table-1 levels (mainly the 128/158-feature subset and reduced budget). And we closed the loop: OOS-robust + parsimony-penalized selection (Result 4) repairs the CSI300 IC degradation (**+0.0007** raw robust vs baseline), while per-date orthogonalization shows a smaller but positive CSI500 residual lift (**+0.0007**; raw robust is **+0.0017**). CSI300 orthogonalization is too aggressive in this run (**-0.0035**), so V3's lesson is universe-aware gating, not blind residualization. Result 6 then exercises the final protocol items on the same run: a portfolio-aware objective re-ranks toward lower-turnover, regime-stable candidates; the prompt/parser contract finds no hidden knobs in the refreshed best Kimi factor; and plateau-aware early-stopping would save 31% of compute at patience=30. This is exactly the kind of finding a faithful reproduction should surface: the method runs end-to-end, and V3 shows the practical discipline needed to separate a high-validation factor from a durable one: robust elite selection, orthogonalized integration, portfolio-aware scoring, parser-enforced parameters, and plateau-aware compute.

Substitutions vs the paper: pure-pandas **Alpha158-128** vs full Alpha158 (exact set needs the qlib package); live configured LLM (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=6`) vs paper Gemini-2.5-Pro; 300-iteration budget (between the paper's 200/400 settings, not the full 400-run grid); report-bootstrapping (FE-report) out of scope → FE-alpha variant. Data IS real (Qlib `.bin` bundle, 2008–2024). See `REPRODUCTION_PLAN.md` / `REAL_DATA.md`.